

Excitation-Aware On-Track System Identification at Full Scale: A Simulation Study

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Abstract

On-track system identification learns tire models from ordinary racing laps instead of dedicated manoeuvres. We study one such residual-learning pipeline at full vehicle scale, inside a simulator that publishes per-wheel forces as ground truth. Transplanted unchanged from a 1:10 platform, it converges on a tire with less than half the vehicle's grip ceiling and most Pacejka coefficients resting on their bounds. The cause is a loss of excitation created by the pipeline's own virtual-data step: the friction the synthetic rollout can demand is a property of the rollout configuration and not of the tire, as a grid on a known tire confirms. Two further defects share that source. The objective the coefficients are fitted against depends on them only through one product, so the peak factor is not recoverable from the rollout at any excitation, and the loop that feeds each fit its own previous answer is not a contraction. We correct all three, and gate the controller interface on derived physical quantities rather than on coefficient bounds. On a static surface 33% below the one the tire prior was measured on, axle cornering stiffness is recovered to within 3.2% and the peak factor to 11%. On a surface losing half its grip under the moving vehicle, the corrected pipeline tracks the decay rate to 7% and completes three timed laps at unchanged lateral tracking error for 18% of lap time, where the inherited configuration holds a grip ceiling nearly four times the plant's and leaves the track.

Keywords: System identification, Tire models, Vehicle dynamics, Autonomous racing, Model predictive control, Persistent excitation

Acknowledgements. The ON-TRACK-SYSID pipeline studied here originates from the ForzaETH Race Stack and the work of Dikici *et al.*, released under an open-source licence. This paper reports a port, a correction and an extension of that work rather than a reimplementation from scratch.

Author Contributions. Conceptualization: Ebrahim Abdelghfar Ebrahim, Mohamed Abdelaziz; Methodology: Ebrahim Abdelghfar Ebrahim, Hazem Abuelanin; Software and investigation: Ebrahim Abdelghfar Ebrahim; Writing — original draft: Ebrahim Abdelghfar Ebrahim; Writing — review and editing: all authors; Supervision: Mohamed Abdelaziz, Mohamed Omar Mohamed Elsayed, Sherif Said. All authors read and approved the final manuscript.

Funding. No funding was received to assist with the preparation of this manuscript.

Declarations

- **Data Availability** Every figure and table of the results section is exported together with the CSV file holding its samples. The scenario definitions, the exported result CSV files, and the figure and sweep generators are released in the project repository, https://github.com/ebrahimabdelghfar/Ebrahim_Master_Thesis_repo.
- **Code Availability** The identification package extends the released baseline of Dikici *et al.* (<https://github.com/ForzaETH/On-Track-SysID>) and is distributed under the MIT licence that upstream’s package manifest declares. The simulator bridge, CARLA_ASU_BRIDGE (https://github.com/ebrahimabdelghfar/Carla_ASU_Bridge), is GPL-3.0. The two run as separate ROS 2 overlays communicating only over ROS 2 topics and services, and neither links against the other, so no combined-work obligation arises between them. The MPC and benchmarking packages are released with the project repository above.
- **Conflict of Interest** The authors declare no competing interests relevant to the content of this article.
- **Ethics approval** Not applicable. All results in this article are obtained in simulation. No experiment involved human subjects, personal data, live vertebrates, or the operation of a physical vehicle.